JATIN SATYAM

Tempe, AZ · Open to relocation | jtnsatyam@gmail.com | linkedin.com/in/jatin-satyam | jatinsatyam.in

EDUCATION

M.S. Robotics & Autonomous Systems (Systems Eng.), Arizona State University · GPA 3.67/4.0 Expected May 2027

B.E. Mechanical Engineering with Major (Honors) in Electric Vehicles, University of Mumbai 2020 – 2024

EXPERIENCE

Research Assistant Aug 2026 – Present

Robotics Systems Instructional Lab, Arizona State University

- Built and commissioned a second bench of a two-robot semiconductor wafer-handling cell (Dobot M1 Pro SCARA, myCobot Pro 600, conveyor, Micro850 PLC), and turned both benches into labs for an ASU short-course manual
- Used the cell's NVIDIA Isaac Sim digital twin to replay real robot motion and find 2 interlock changes that cut the real cell's cycle time 26% (139.9 to 103.9 s, reproduced in 3 of 3 runs)

Mechatronics Engineer Jan 2025 – Jul 2025

Paras Defence & Space Technologies Ltd., Navi Mumbai, India

- Designed and iterated Gifford-McMahon cryocooler mechanisms in SolidWorks and NX, refining moving-assembly tolerances and interfaces for reliable cold-head displacer motion and manufacturability
- Led performance testing of 3 cryogenic cooling assemblies across 25+ instrumented runs, characterizing thermal losses and translating findings into mechanism-level design changes that cut prototype rework 12%

CFD Engineer Aug 2024 – Nov 2024

MARS Exploration Pvt. Ltd, Remote

- Performed 3D CFD analysis of NACA 23012 and NACA 4412 airfoils in ANSYS Fluent 2024 R2 (10-degree AoA, 48 m/s) - geometry, boundary-layer-refined meshing with quality checks, solver setup, and convergence monitoring

R&D Engineer May 2023 – Aug 2023

TriQual Cryogenics Pvt Ltd, Navi Mumbai, India

- Contributed to mechanical design and vibration control for Stirling-cycle cryocoolers operating near 70 K

PROJECTS

Wafer-Cell Digital Twin ↗ *· NVIDIA Isaac Sim 6.1 · ROS 2 Jazzy · PyTorch* 2026

- Live twin of a real two-robot cell in Isaac Sim and Gazebo over a read-only ROS 2 bridge, tracking the real arms within 0.5 / 2.2 degrees; wafer contact physics matched the bench (2.6 mm slide predicted, 2.1-2.5 mm measured)
- Trained a ResNet-18 tape classifier only on synthetic Isaac Sim images (92.6% on 136 unseen real camera crops) and built motion-based anomaly detection that caught a real cell fault and alarms 5.4 s into a stall

Space Robotics & AI ↗ *· ROS 2 Jazzy · Gazebo Harmonic · PX4 SITL* 2026

- Aerial-ground autonomy on real Mars terrain: a UAV maps HiRISE Jezero Crater terrain into a Bayesian occupancy grid (0.25 m/cell) that a 6-wheel rover navigates with A* replanning; terrain-relative navigation at 2 Hz over K=294 pose candidates
- Also: Drone search, mapping & precision landing (PX4, ArUco, RTAB-Map SLAM; 150/150), Apollo 17 photogrammetry & 3D Gaussian splatting, Cart-pole LQR, Boustrophedon coverage controller

Robot Arm Manipulation Lab ↗ *· Python · OpenCV · Dobot Magician Lite* 2025

- Natural-language manipulation with vision & LLM control on a physical Dobot Magician Lite: Llama 3.3 70B on Groq parses commands, HSV vision locates objects, validated motion primitives execute; camera-to-robot calibration to +/-2 mm

Vision-Guided Aerial Autonomy ↗ *· Simulink · Flight Control · Machine Vision* 2026

- Controlled landing on a moving platform: Simulink controllers deployed to a physical quadrotor track a platform sliding along a track from above and land on it in motion, in one continuous take (position held within ~5 cm of the platform in simulation)

SKILLS

Programming: Python (NumPy, SciPy, OpenCV, Open3D), C++, C, MATLAB, Bash, SQL

Software: ROS 2 (Humble/Jazzy), Gazebo, NVIDIA Isaac Sim, Omniverse Replicator, PyTorch, ONNX, PX4 SITL, MuJoCo, Simulink, RTAB-Map, Docker, SolidWorks, Siemens NX, ANSYS, Git & GitHub

Other Skills: Kinematics, Motion Planning, A* Planning, SLAM, Occupancy Mapping, PID, LQR, State-Space Modeling, Synthetic Data, Anomaly Detection, PLC (Micro850), Modbus TCP

AWARDS & RECOGNITION

- 2nd Prize, National Level Project Poster Competition, IEI-FCRIT 2024 - gyro motor rotor (CALIBRE 2k24)
- Winner, MLH Best Use of Auth0 - Meridian, Innovation Hacks 2.0, ASU, Apr 2026
- Co-inventor, invention disclosure filed with ASU's Skysong Innovations - SipSafe Smart Cup, 2026